Final Report

Implementation of Keren (2003) - Recognizing Human Activities in Video Using Local Features and Naive Bayes

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1. Summary of the Paper

The paper "Recognizing image style and activities in video using local features and naive Bayes" by Daniel Keren (Pattern Recognition Letters, 2003) addresses the problem of recognizing human activities in video sequences using classical signal processing and probabilistic classification.

The core idea is to represent short spatio-temporal regions of a video as compact numerical descriptors using the 3D Discrete Cosine Transform (DCT). The video is divided into small overlapping 5x5x5 blocks (5 pixels wide, 5 pixels tall, 5 frames deep), each capturing local motion and texture over a short time window. Each block is normalized and transformed via 3D DCT into a 125-dimensional feature vector encoding both spatial and temporal frequency patterns.

Not all 125 coefficients are equally useful. The paper uses Mutual Information (MI) to select the most discriminative features - those whose distribution differs most between the two activity classes. The top selected features are used to train a Naive Bayes classifier, which learns a binary threshold per feature and classifies blocks by accumulating log-probabilities across features.

At inference time, the classifier slides across every pixel position in each frame and labels each block as one of the two activity classes, producing a per-pixel activity map. A confidence filter ensures only high-certainty predictions are displayed - a pixel is labeled only if the winning class scores at least twice the losing class.

2. Implementation

The implementation follows the pipeline described in the paper closely. All training videos were converted from MP4 format to grayscale frames and resized to 64x64 pixels. A sliding window of size 5x5x5 is applied at every pixel position (stride 1) across each frame sequence. Before feature extraction, a motion filter discards any block whose pixel variance falls below a threshold of 20.0. This effectively ignores static background regions and reduces noise.

Each remaining block is normalized to zero mean and unit variance, then transformed using the 3D DCT to produce a 125-dimensional feature vector. Mutual Information is computed between each coefficient and the class label, the DC component is excluded, and the top 10 most informative coefficients are selected for training. A Naive Bayes classifier is trained on a class-balanced dataset (equal number of walking and waving samples) by computing per-feature thresholds and log-probabilities.

At inference time, the same pipeline is applied to the test video. Each pixel position is classified independently, and a confidence filter (winner score >= 2x loser score) ensures only high-certainty detections are displayed. The result is a per-pixel activity map overlaid on the original video - yellow pixels indicate waving, purple pixels indicate walking.

3. Improvements

Several improvements were made beyond the original paper's implementation. Not all attempts were successful. The failed attempts are documented as well, as they informed us of the final design decisions.

3.1 Higher Resolution (128x128)

The most impactful improvement was increasing the frame resolution from 64x64 to 128x128 pixels. This produced approximately 5x more training samples per video, since more 5x5x5 blocks pass the motion filter in higher-resolution frames. The larger frames also contain finer spatial detail, allowing the DCT features to capture more precise motion patterns.

3.2 Data Augmentation (Horizontal Flip)

Each training video was also processed with a horizontal flip, effectively doubling the training dataset. This adds robustness to viewpoint asymmetries. For example, a person walking from left to right vs. right to left produces mirrored motion patterns that the classifier should treat identically.

3.3 Additional Classifiers (SVM and Logistic Regression)

Two additional classifiers were trained on the 128x128 DCT features:

- SVM with RBF kernel (C=1): Finds a curved decision boundary in feature space that maximally separates the two classes. Cross-validation on the training set showed C=1 outperformed C=10 by 2.7%.
- Logistic Regression (L2, C=1): Combines all features into a single weighted sum and outputs a probability score.

Both classifiers use the top 20 MI-selected features and StandardScaler normalization.

3.4 Ensemble Model (Majority Vote)

The three classifiers (Naive Bayes, SVM, Logistic Regression) were combined into an ensemble using majority vote. Each pixel is labeled only if at least 2 of the 3 models agree. This reduces individual classifier errors and produces more reliable predictions.

3.5 Post-Processing

Three post-processing steps were applied to clean up the raw per-pixel predictions:

- Temporal smoothing (5-frame window): For each pixel, a majority vote across 5 consecutive frames eliminates frame-to-frame label flicker.
- Blob cleanup (minimum 60px): Connected colored regions smaller than 60 pixels are removed, eliminating isolated noise specks.
- Dominant-class suppression (45% threshold): If waving pixels account for less than 45% of a frame's labeled pixels, all waving labels in that frame are erased, and vice versa. This reduces cross-contamination between activity sections.

3.6 Things That Were Tried But Did Not Work

Not every approach improved the results. The following were attempted and either reverted or abandoned:

- Deep learning model: A deep learning-based classifier was considered at the start of the project but was too computationally heavy to run locally in reasonable time. The project was therefore scoped to classical methods only.
- Histogram of Optical Flow (HOF): HOF features were tested as an alternative to DCT. While optical flow captures motion direction explicitly, the results were not as clean as DCT-based features for this dataset and were abandoned.
- Larger block size (7x7x7): Increasing the block size from 5x5x5 to 7x7x7 produced worse visual results. The larger blocks blurred the spatial boundaries between activity regions. Reverted to 5x5x5.
- More MI features (top 30): Increasing the number of selected features from 20 to 30 had negligible effect. The additional features carried little discriminative information.
- Larger temporal smoothing window (15 frames): Made results worse, as the larger window blended the walking and waving sections of the video together near the transition point.
- Larger blob size threshold (150px): Had no visible effect. The wrong-class blobs in the test video were already larger than 150 pixels.
- Logistic Regression C tuning: Testing different C values produced completely flat results. The model was insensitive to regularization strength for this feature set.

Summary of all improvements:

Improvement	Effect
128x128 resolution	Major - 5x more training data
Data augmentation (horizontal flip)	Increased robustness
SVM (RBF, C=1)	+2.7% on training cross-validation
Logistic Regression (L2, C=1)	Added classifier diversity
Temporal smoothing	Eliminated label flicker
Dominant-class suppression	Reduced cross-contamination
7x7x7 block size	Worse - reverted
Top 30 MI features	No improvement - reverted
HOF features	Not successful - abandoned
Deep learning	Too heavy for local execution - abandoned

4. Results & Visualizations

The system was evaluated on a single test video (test.mp4) containing both a walking section and a waving section. Since no ground truth per-pixel annotations exist for the test video, evaluation was done visually by inspecting the output videos.

4.1 Original Model (DCT + Naive Bayes)

The original model recognized the activities reasonably well, but with several visible limitations. The labeled pixels were patchy rather than covering the full body. Only specific regions with strong local motion were colored at any given frame. There was also noticeable noise: yellow (waving) patches occasionally appeared during the walking section due to arm swing creating similar DCT frequency patterns to waving, and the colored labels flickered from frame to frame rather than remaining stable.

Figure 1: Original model - walking (purple pixels on legs + noise)

Figure 2: Original model - waving (yellow pixels on arms)

4.2 Ensemble Model (NB + SVM + LogReg)

The ensemble model produced noticeably better results. Recognition was more reliable, with significantly less noise - fewer wrong-color patches appeared during each section. The temporal smoothing eliminated the flickering, producing stable labels across frames. The colored region covered almost the entire body rather than isolated patches, giving a much cleaner and more complete detection.

Figure 3: Ensemble model - walking (purple pixels covering body)

Figure 4: Ensemble model - waving (yellow pixels covering more of the motion)

4.3 Comparison

	Original Model	**Ensemble Model**
Resolution	64x64	128x128
Classifiers	Naive Bayes only	NB + SVM + LogReg
Localization	Patchy - specific body parts only	Nearly full body coverage
Noise	Noticeable - wrong-color patches	Much reduced
Temporal stability	Flickering labels	Stable across frames
Post-processing	Confidence filter only	Full post-processing pipeline

4.4 Limitations

- The labeled regions do not cover the full body in the original model - only blocks with strong local motion get classified, leading to a patchy appearance.
- During the walking section in the original model, arm swing occasionally produces yellow (waving) patches - arm swing and waving create similar DCT frequency patterns.
- Neither model was evaluated with formal metrics (accuracy, precision, recall, F1) due to the absence of ground truth pixel-level annotations for the test video.

5. Discussion & Findings

5.1 What Worked Well

The most significant finding was that increasing the resolution from 64x64 to 128x128 had a larger impact than any other single change. It not only provided more training data, but also

gave the DCT features finer spatial detail to work with. This suggests that for local feature-based methods, the amount and quality of training data matters more than classifier complexity.

The post-processing pipeline was equally important. Without temporal smoothing and dominant-class suppression, the raw classifier output was too. This highlights a common challenge in video understanding: even a good per-frame classifier needs temporal context to produce coherent results.

The ensemble approach added robustness. By requiring agreement from multiple classifiers, it reduced false detections. However, it did not dramatically change the visual output compared to using a single well-tuned classifier on 128x128 features. This suggests the three classifiers were making similar errors, likely because they all operated on the same DCT features.

5.2 What Did Not Work and Why

Several approaches failed to improve results. Larger block sizes (7x7x7) blurred spatial boundaries. More MI features (top 30) added noise rather than information. Larger temporal smoothing windows blended the two activity sections together near the transition. These failures share a common theme: adding more complexity to a simple, well-calibrated system often hurts rather than helps.

The HOF features were less effective than DCT for this dataset, likely because optical flow estimation itself introduces errors on the relatively low-quality training videos.

5.3 Fundamental Limitations of the Approach

The DCT local feature approach has inherent limitations that no amount of tuning can fully overcome:

- It captures local patterns in small 5x5x5 regions. It has no understanding of the global body structure or long-term motion sequences.
- Similar local motions (arm swing vs. waving) can produce similar DCT signatures, making them hard to distinguish at the block level.
- The method is sensitive to camera movement and lighting changes, since these affect pixel intensity patterns directly.

Modern deep learning approaches, such as CNN-based action recognition or transformer models, address these limitations by learning hierarchical features over the entire frame and long temporal sequences. However, they require orders of magnitude more data and computational resources, making classical methods like this still relevant for constrained, low-compute scenarios.

6. References

1. Keren, D. (2003). Recognizing image style and activities in video using local features and naive Bayes. Pattern Recognition Letters, 24(15), 2913-2922.

2. AI assistance log: docs/ai_assistance_log.md